

A deep-field astronomical image showing a vast field of galaxies and stars against a black background. The galaxies are in various colors (blue, orange, white) and shapes (spiral, elliptical, irregular). Two thin blue horizontal lines are positioned above and below the central text.

ARITHMETIC

A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a black space. Two horizontal blue lines frame the central text.

ADDITION

+ Base-16		2	A
	4	B	F

.

.

.

E	5	4
2	B	

PADDING

+ Base-16	0	2	A
	4	B	F

.
.
.

E	5	4
2	B	0

+ Base-16	0	2	A
	4	B	F

.

.

.

E	5	4
2	B	0
		4

+ Base-16	0	2	A
	4	B	F

.

.

.

E	5	4
2	B=11	0
	16	4

$$\frac{16}{16} = 1 \text{ } r \text{ } 0$$

+ Base-16	0	2	A
	4	B	F

.	1		
	E	5	4
	2	B	0
		0	4

$$\frac{16}{16} = 1 \text{ } r \text{ } 0$$

+ Base-16	0	2	A
	4	B	F

.	1		
	E=14	5	4
	2	B	0
.	17	0	4

$$\frac{17}{16} = 1 \text{ r } 1$$

			1
+	0	2	A
Base-16	4	B	F

.

.

.

1		
E	5	4
2	B	0
1	0	4

$$\frac{17}{16} = 1 \text{ r } 1$$

			1
+ Base-16	0	2	A=10
	4	B	F=15
			26

.

.

.

1		
E	5	4
2	B	0
1	0	4

$$\frac{26}{16} = 1 \text{ } r \text{ } 10$$

		1	1
+ Base-16	0	2	A
	4	B	F
			10

.
.
.

1		
E	5	4
2	B	0
1	0	4

$$\frac{26}{16} = 1 \text{ } r \text{ } 10$$

		1	1
+ Base-16	0	2	A
	4	B	F
			A

.

.

.

1		
E	5	4
2	B	0
1	0	4

		1	1
+ Base-16	0	2	A
	4	B=11	F
		14	A

.

.

.

1		
E	5	4
2	B	0
1	0	4

		1	1
+ Base-16	0	2	A
	4	B	F
		E	A

.

.

.

1		
E	5	4
2	B	0
1	0	4

		1	1
+ Base-16	0	2	A
	4	B	F
	4	E	A

.

.

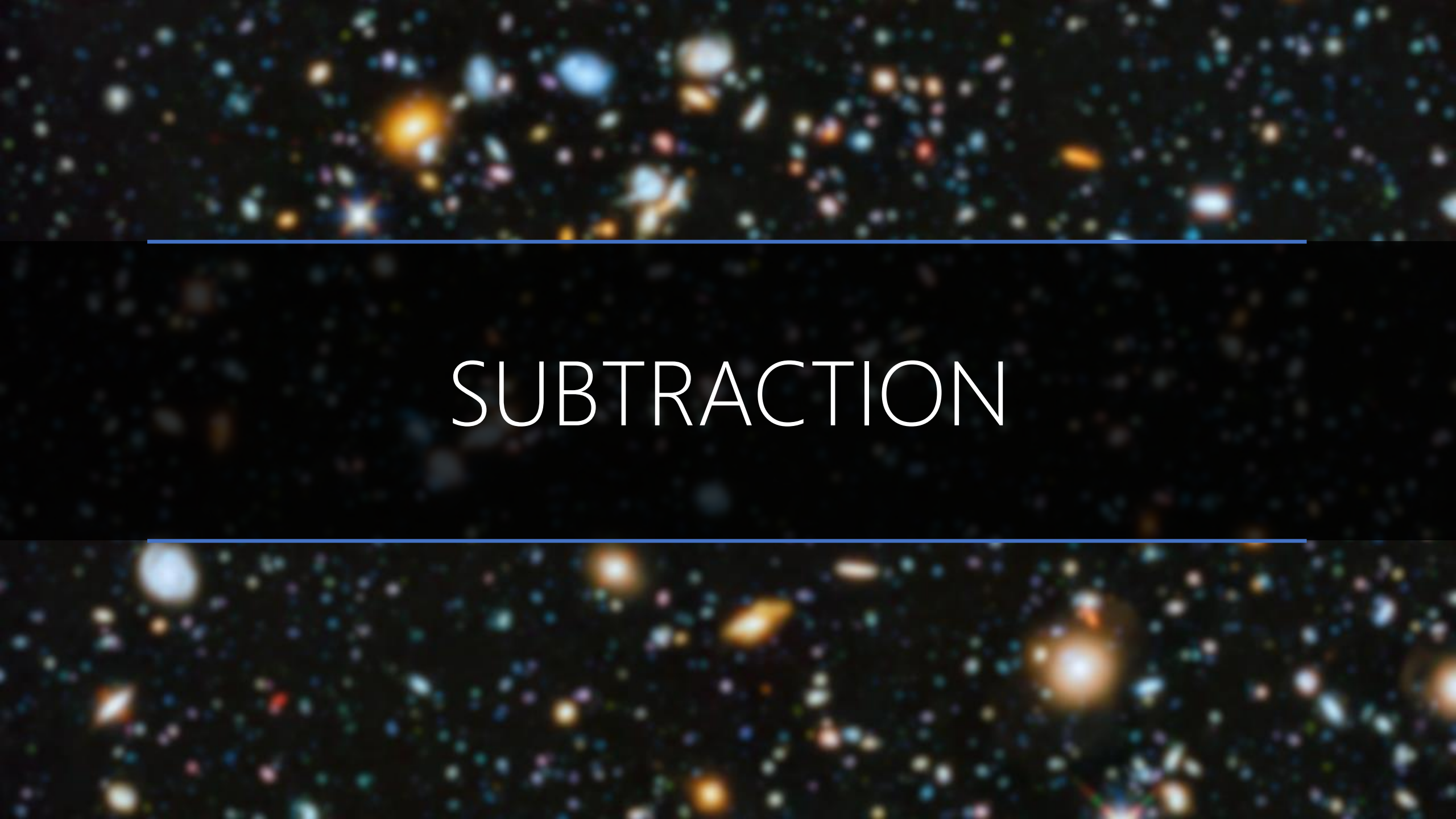
.

1		
E	5	4
2	B	0
1	0	4

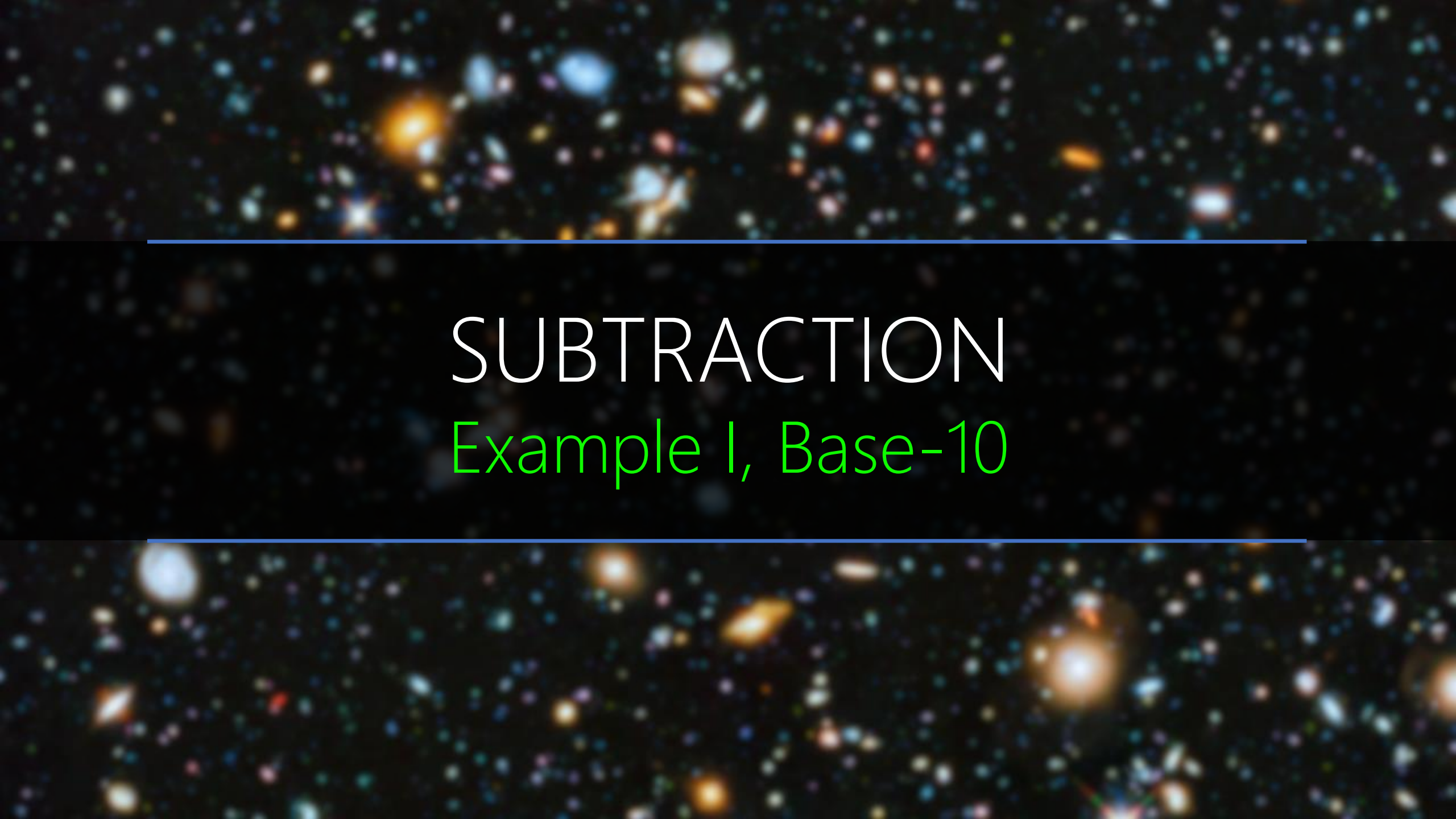
The background of the entire slide is a deep space image showing a vast field of galaxies and stars. The galaxies are in various stages of evolution, some appearing as bright, fuzzy blobs and others as more defined structures. The colors range from warm oranges and yellows to cool blues and purples. Two thin, horizontal blue lines are positioned above and below the central text, spanning most of the width of the image.

MULTIPLY

At Home

The background of the slide is a deep space image showing a dense field of galaxies and stars. The galaxies are in various stages of evolution, with some appearing as bright, irregular blobs and others as more structured, elongated shapes. The stars are small, bright points of light scattered across the dark background. The overall color palette is dominated by the blues and purples of the cosmic background, with the warm colors of the galaxies and stars providing contrast.

SUBTRACTION

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines frame the central text.

SUBTRACTION

Example I, Base-10

— Base-10		2	1
	4	3	2

.

.

.

1	5	4
0	6	

PADDING

— Base-10	0	2	1
	4	3	2

.
.
.

1	5	4
0	6	0

IS IT POSSIBLE?

— Base-10	0	2	1
	4	3	2

.

.

.

1	5	4
0	6	0

— Base-10	0	2	1
	4	3	2

.

.

.

1	5	4
0	6	0
		4

— Base-10	0	2	1
	4	3	2

.

.

.

1	5	4
0	6	0
		4

				BORROW	
—	0	2	1	.	
Base-10	4	3	2	.	
				.	

-1	→ +10	
1	5	4
0	6	0
	= 10+5-6 = 9	4

— Base-10	0	2	1
	4	3	2

.	-1	+10	
.	1	5	4
.	0	6	0
.		9	4

— Base-10	0	2	1
	4	3	2

.	-1	+10	
.	1	5	4
.	0	6	0
.	0	9	4

— Base-10	0	2	1
	4	3	2

.

.

.

-1	+10	
1	5	4
0	6	0
0	9	4

		-1	+10
— Base-10	0	2	1
	4	3	2
			9

.

.

.

-1	+10	
1	5	4
0	6	0
0	9	4

		-1	+10
— Base-10	0	2	1
	4	3	2
			9

.

.

.

-1	+10	
1	5	4
0	6	0
0	9	4

		+10	
	-1	-1	+10
—	0	2	1
Base-10	4	3	2
		8	9

.

.

.

-1	+10	
1	5	4
0	6	0
0	9	4

	+10	+10	
-1	-1	-1	+10
— Base-10	0	2	1
	4	3	2
	5	8	9

.

.

.

-1	+10	
1	5	4
0	6	0
0	9	4

	+10	+10	
-1	-1	-1	+10
—	0	2	1
Base-10	4	3	2
	5	8	9

.

.

.

-1	+10	
1	5	4
0	6	0
0	9	4

Last Borrow → 021.154 < 432.060 → Negative Result

	+10	+10	
-1	-1	-1	+10
— Base-10	0	2	1
	4	3	2
	5	8	9

.

.

.

-1	+10	
1	5	4
0	6	0
0	9	4

The Result is not correct.

	+10	+10	
-1	-1	-1	+10
— Base-10	0	2	1
	4	3	2
	5	8	9

.

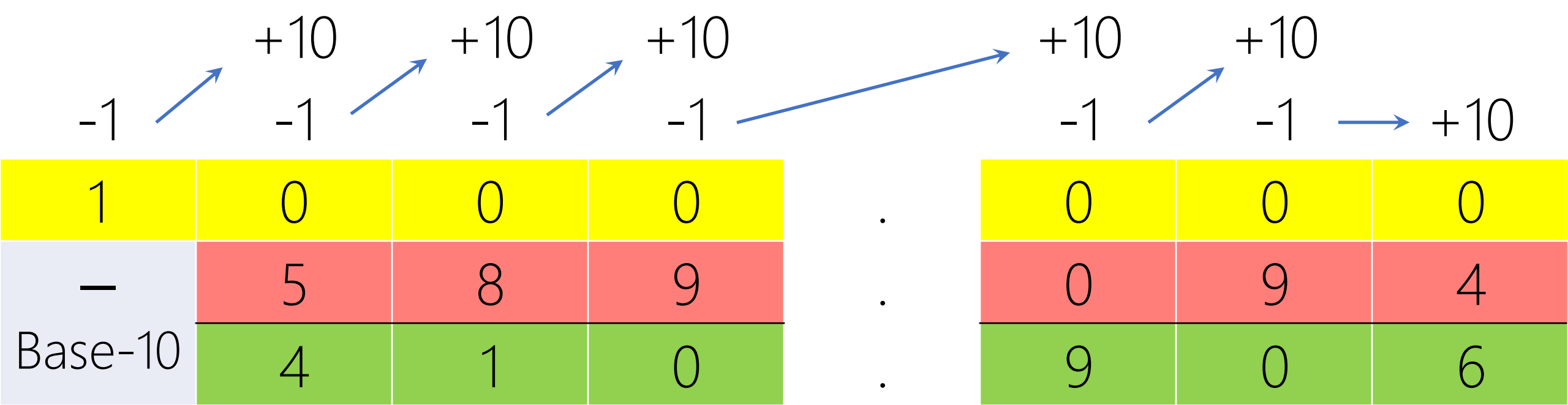
.

.

-1	+10	
1	5	4
0	6	0
0	9	4

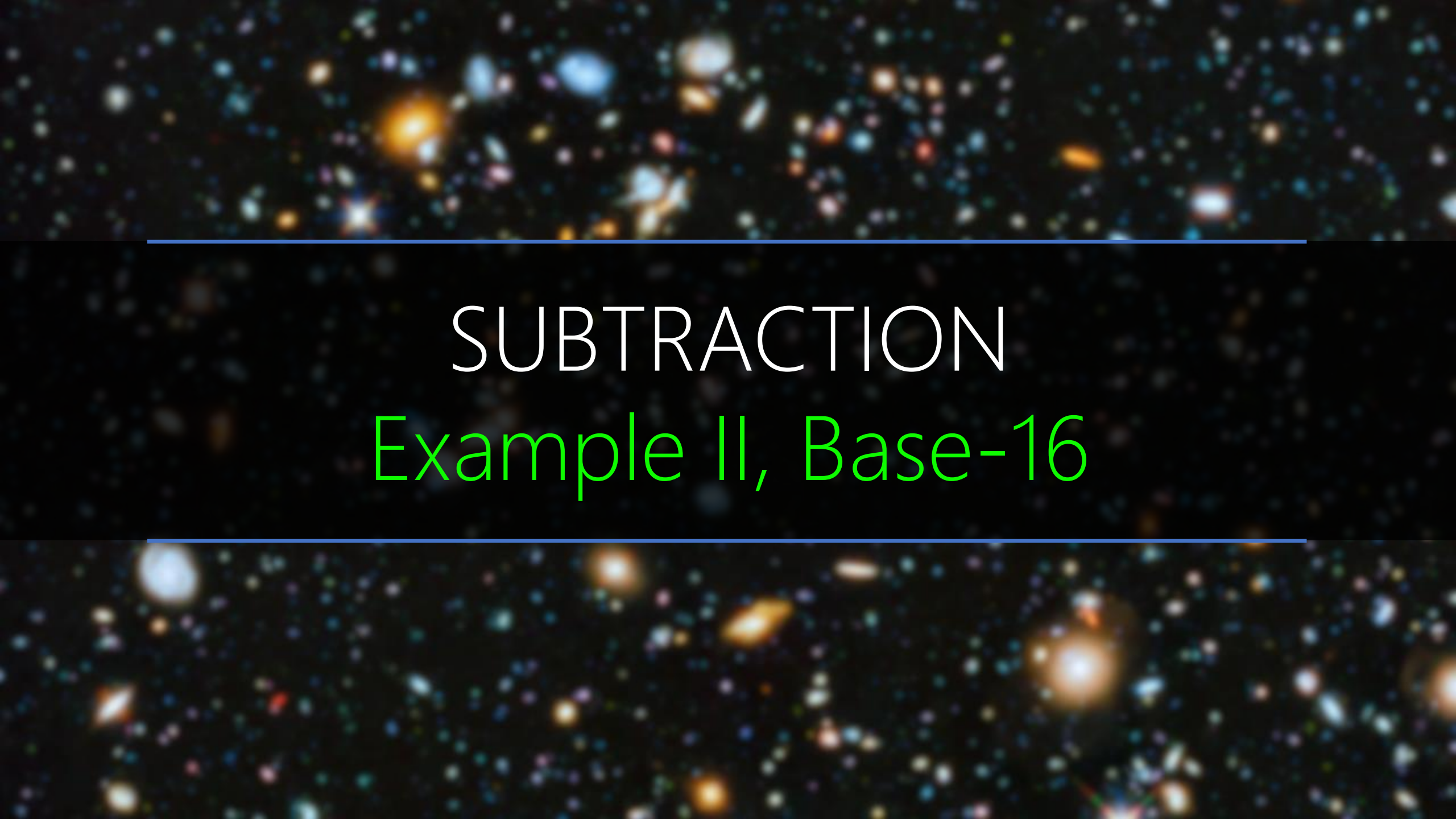
$$\begin{aligned}
 &= (021.154)_{10} - (432.060)_{10} = (021.154)_{10} + (1000.000)_{10} - (1000.000)_{10} - (432.060)_{10} \\
 &= - (1000.000)_{10} + (589.094)_{10} = - [(1000.000)_{10} - (589.094)_{10}]
 \end{aligned}$$

Extra Subtraction:



Equation showing the subtraction process:

$$\begin{aligned} &= (021.154)_{10} - (432.060)_{10} = (021.154)_{10} + (1000.000)_{10} - (1000.000)_{10} - (432.060)_{10} \\ &= - (1000.000)_{10} + (589.094)_{10} = - [(1000.000)_{10} - (589.094)_{10}] \\ &= - (410.906)_{10} \end{aligned}$$

The background of the slide is a deep space image showing a dense field of galaxies. These galaxies appear as bright, colorful spots in various colors including orange, yellow, blue, and white against a black background. Two thin, horizontal blue lines are positioned above and below the central text.

SUBTRACTION

Example II, Base-16

— Base-16		2	A
	4	B	F

.

.

.

E	5	4
2	B	

PADDING

— Base-16	0	2	A
	4	B	F

.
.
.

E	5	4
2	B	0

— Base-16	0	2	A
	4	B	F

.

.

.

E	5	4
2	B	0
		4

— Base-16	0	2	A
	4	B	F

.

.

.

E	5	4
2	B=11	0
		4

				BORROW	
—	0	2	A	.	
Base-16	4	B	F	.	
				.	

-1	→ +16	
E	5	4
2	B=11	0
	= 5+16-11 = 10 = A	4

— Base-16	0	2	A
	4	B	F

.	-1	+16	
.	E	5	4
.	2	B	0
.		A	4

— Base-16	0	2	A
	4	B	F

.	-1	+16	
.	E=14	5	4
.	2	B	0
.	B=11	A	4

— Base-16	0	2	A=10
	4	B	F=15

.

.

.

-1	+16	
E	5	4
2	B	0
B	A	4

		-1	+16
— Base-16	0	2	A=10
	4	B	F=15
			B=11

.

.

.

-1	+16	
E	5	4
2	B	0
B	A	4

		-1	+16
— Base-16	0	2	A
	4	B=11	F
			B

.

.

.

-1	+16	
E	5	4
2	B	0
B	A	4

		+16	
	-1	-1	+16
—	0	2	A
Base-16	4	B=11	F
		6	B

.

.

.

-1	+16	
E	5	4
2	B	0
B	A	4

		+16	
	-1	-1	+16
—	0	2	A
Base-16	4	B	F
		6	B

.

.

.

-1	+16	
E	5	4
2	B	0
B	A	4

	+16	+16	
-1	-1	-1	+16
— Base-16	0	2	A
	4	B	F
	B=11	6	B

.

.

.

-1	+16	
E	5	4
2	B	0
B	A	4

	+16	+16	
-1	-1	-1	+16
—	0	2	A
Base-16	4	B	F
	B	6	B

.

.

.

-1	+16	
E	5	4
2	B	0
B	A	4

02A.E54 < 4BF.2B0

Last Borrow → Negative Result

	+16	+16	
-1	-1	-1	+16
— Base-16	0	2	A
	4	B	F
	B	6	B

.

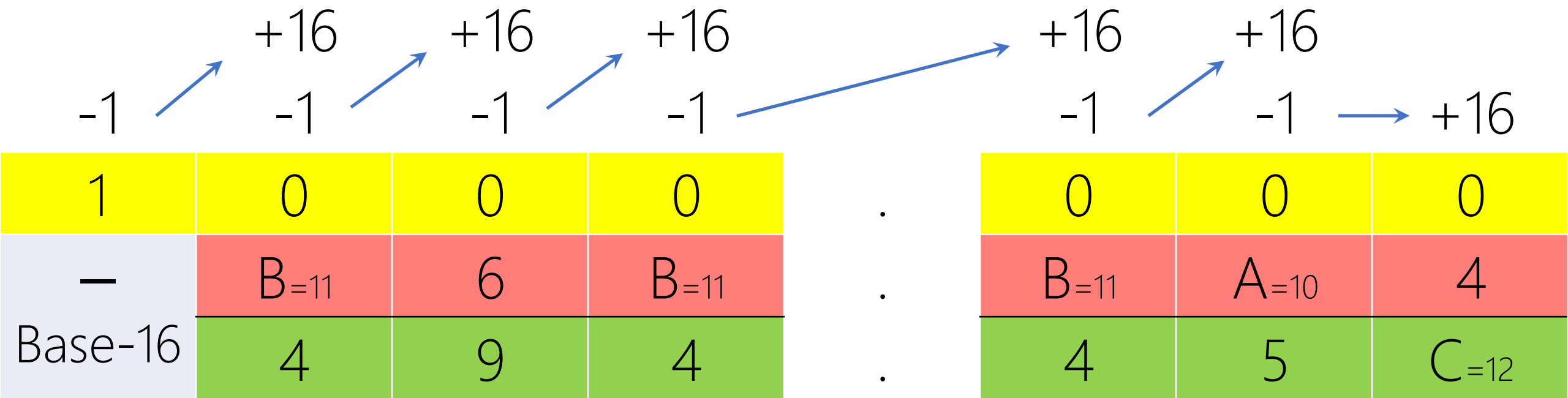
.

.

-1	+16	
E	5	4
2	B	0
B	A	4

$$\begin{aligned}
 &= (2A.E54)_{16} - (4BF.2B0)_{16} = (2A.E54)_{16} + (1000.000)_{16} - (1000.000)_{16} - (4BF.2B0)_{16} \\
 &= -(1000.000)_{16} + (B6B.BA4)_{16} = -[(1000.000)_{16} - (B6B.BA4)_{16}]
 \end{aligned}$$

Extra Subtraction:



Mathematical derivation of the result:

$$\begin{aligned} &= (2A.E54)_{16} - (4BF.2B0)_{16} = (2A.E54)_{16} + (1000.000)_{16} - (1000.000)_{16} - (4BF.2B0)_{16} \\ &= -(1000.000)_{16} + (B6B.BA4)_{16} = -[(1000.000)_{16} - (B6B.BA4)_{16}] \\ &= -(494.45C)_{16} \end{aligned}$$

A cosmic background image featuring a dense field of galaxies and stars. The galaxies are in various colors, including blue, orange, and white, against a dark, star-filled sky. Two horizontal blue lines are positioned above and below the central text.

DIVISION

At Home

A cosmic background image featuring a dense field of galaxies and stars. The galaxies are in various stages of evolution, with some appearing as bright, irregular shapes and others as more structured, spiral or elliptical forms. The colors range from deep blues and purples to bright yellows and oranges, set against a dark, star-filled sky.

SIGNED NUMBERS